

# John Girgis

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## EDUCATION

### New Jersey Institute of Technology, Albert Dorman Honors College

Anticipated Grad. 2028

*B.S. in Computer Engineering, GPA: 4.0*

*Newark, NJ*

**Awards:** Donald K. Blackman ECE IAB Academic Excellence Award, Honors Full Merit Scholarship, High School Class of 2024 Salutatorian

**Coursework:** Analytical & Computational Neuroscience, Linear Algebra, Honors Differential Equations

## EXPERIENCE

### Undergraduate Researcher

Jun 2026 – Present

*Department of Neuroscience, Yale School of Medicine, Yale University*

*New Haven, CT*

- Built statistical models to predict gaze targeting from video motion features during naturalistic viewing
- Fit linear dynamical systems to neural time-series data and applied eigendecomposition to characterize interpretable latent dynamics.
- Developing biologically constrained RNNs to model the neural pathways underlying visual decision-making
- Presented findings at the Undergraduate Research Institute (URI) Symposium

### Undergraduate Research Assistant

Jan 2025 – Present

*Princeton Neuroscience Institute, Princeton University*

*Princeton, NJ*

#### **Buschman Lab** (*Sept. 2025 - Present*)

- Designed a variational autoencoder to learn latent representations of visual processing underlying saccadic prediction
- Implemented a PyTorch feedforward network to predict latent saccades
- Built a parameterized Blender pipeline to generate synthetic retinal datasets for model development and evaluation

#### **Engel Lab** (*Jan. 2025 - May 2025*)

- Analyzed RNN architectures and used Principal Component Analysis (PCA) visualizations to evaluate neural compositionality and generalization
- Designed PyTorch RNNs to study compositionality in structurally modulated neuron populations

### Undergraduate Researcher

Feb 2025 – Oct 2025

*Department of Data Science, New Jersey Institute of Technology*

*Newark, NJ*

- Developed an end-to-end PyTorch pipeline for brain age prediction from MEG data, with applications to neurodegenerative disease classification
- Developed a Hyperbolic Graph Convolutional Network (GCN) that outperformed conventional GCN and classical baselines across binary and multiclass classification tasks
- Presented findings at the 2025 IEEE MIT Undergraduate Research Technology Conference

### Experiential Learning Officer, IEEE NJIT

Oct 2024 – Present

*New Jersey Institute of Technology*

*Newark, NJ*

- Designed a project based AI curriculum used by over 125 recurring users spanning deep learning, reinforcement learning, computer vision, NLP, RAG, and agentic AI
- Wrote and curated 10 Google Colab notebooks and facilitated more than 30 weekly technical sessions

## PUBLICATIONS

**Girgis, J. & Xu, M. (2025).** “End-to-End Hyperbolic Graph Neural Networks for Brain Age Prediction with MEG Data.” *2025 IEEE MIT Undergraduate Research Technology Conference (URTC)*. doi: [10.1109/URTC68753.2025.11533033](https://doi.org/10.1109/URTC68753.2025.11533033)

## PROJECTS

### Memento | HackMIT '25 | MentraOS, Python, Flask, Tesseract

Sep 2025

- Built a smart glasses networking assistant in 24 hours using React Native, Flask, and Python based vision pipelines
- Integrated Tesseract OCR, facial recognition, live callbacks, and cloud-backed processing to deliver real time contact capture and user feedback

### Cera One | PCB, CAD, Python, MQTT

Sep 2025 – Present

- Developing a dry sEMG based wearable band to add accessibility control to daily use devices

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C++, Swift, JavaScript, HTML/CSS, Matlab, Lua

**Libraries:** Pandas, NumPy, Matplotlib, PyTorch, OpenCV

**Frameworks:** React, Node.js, Flask, Material-UI, TailwindCSS

**Engineering Tools:** Fusion360, Autodesk Inventor, Onshape, KiCAD, Multisim